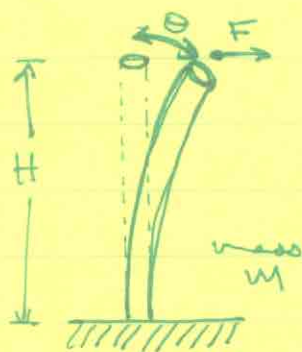


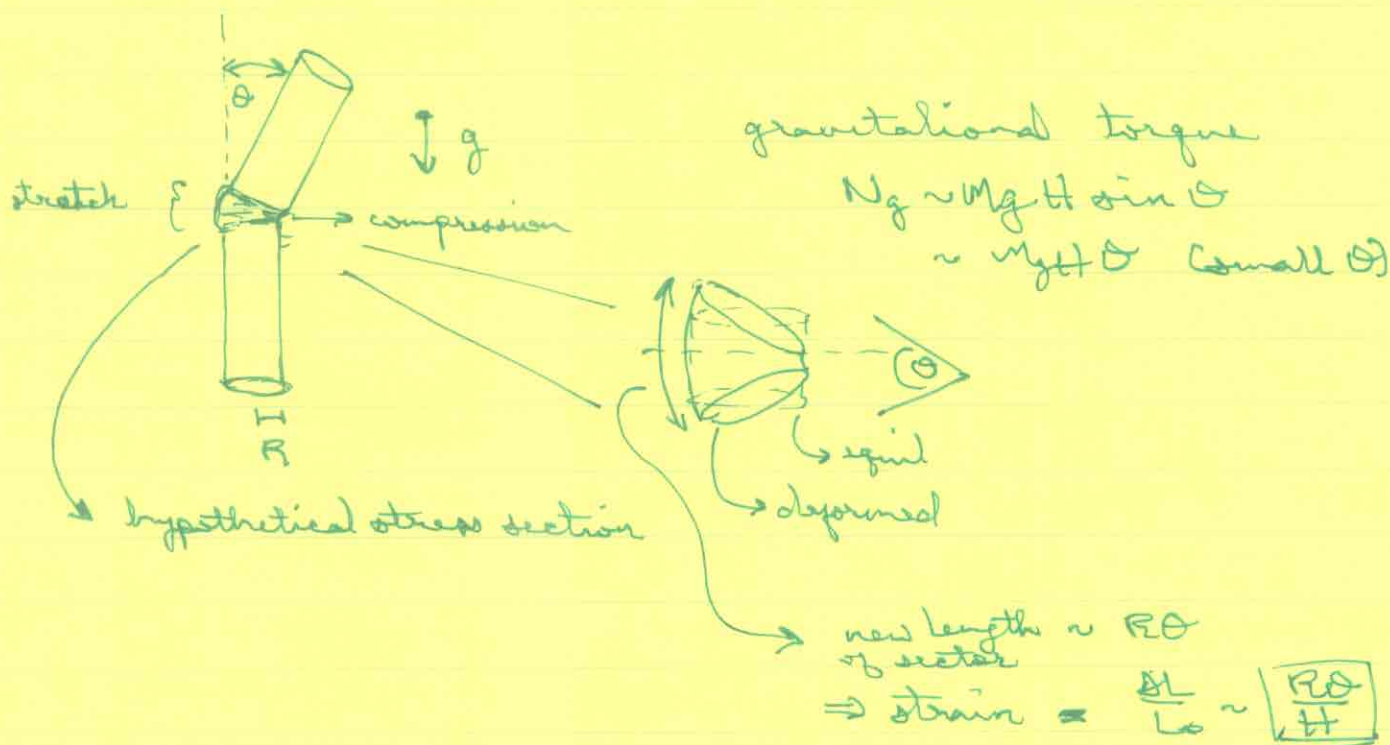
Height Limit for a Vertical Elastic Beam



$$E = \frac{\text{stress}}{\text{strain}} \rightarrow \text{Young's modulus}$$

(measurable property of material)

Consider the more idealized configuration :



The gravitational torque produces the force on the sector; its magnitude is given by

$$\underbrace{F_s \cdot R}_{\text{torque transmitted to sector}} \sim N_g \Rightarrow F_s \sim \frac{mgH\theta}{R}$$



$$\Rightarrow \text{stress on joint} \sim \frac{F_s}{R^2} \sim \frac{mgH\theta}{R^3}.$$

In equilibrium the stresses on the joint are balanced by its internal elastic press, so that

$$\epsilon \sim \frac{(mgH\theta/R^3)/(R\theta/H)}{R^4} = \frac{mgH^2}{R^4}.$$

The mass of the beam can be written as

$$m \sim \rho \cdot R^2 H \Rightarrow \frac{mgH^2}{R^4} \sim \frac{\rho g H^3}{R^2}$$

$$\Rightarrow (\text{equilibrium}) \quad \epsilon \sim \frac{\rho g H^3}{R^2}.$$

When H is larger than $\sim \left(\frac{\epsilon R^2}{\rho g}\right)^{1/3}$ the gravitational torque overpowers the compensating elastic response, and there can be no equilibrium — the beam topples. So,

$$H_{\max} \sim \left(\frac{\epsilon R^2}{\rho g}\right)^{1/3} \quad \text{or} \quad \boxed{\left(\frac{\epsilon A_{\text{cross}}}{\rho g}\right)^{1/3}}.$$